Qiskit Fall Fest 2026

Challenge 3: Can You Change the Correlation? — Complete Solution

As required by the uploaded PDF (page 1), this solution includes the original Bell circuit, a one-gate modification, predictions, 100-shot execution code, histograms, and an explanation.

1. Original Bell-state circuit

Prediction: H on qubit 0 followed by CNOT (control 0, target 1) prepares (|00■ + |11■)/√2. Ideally, only 00 and 11 occur, each with probability 50%.

```
from qiskit import QuantumCircuit, transpile
from qiskit_aer import AerSimulator
from qiskit.visualization import plot_histogram
import matplotlib.pyplot as plt

simulator = AerSimulator()
bell = QuantumCircuit(2, 2)
bell.h(0)
bell.cx(0, 1)
bell.measure([0, 1], [0, 1])
result = simulator.run(transpile(bell, simulator), shots=100).result()
counts_bell = result.get_counts()
print(counts_bell)
plot_histogram(counts_bell); plt.show()
```

2. Modified circuit — Add one X gate

Modification: Add X to qubit 0 after the Bell-state preparation and before measurement. The state becomes (|10■ + |01■)/√2, so the qubits have opposite outcomes.

```
modified = QuantumCircuit(2, 2)
modified.h(0)
modified.cx(0, 1)
modified.x(0)
modified.measure([0, 1], [0, 1])
result = simulator.run(transpile(modified, simulator), shots=100).result()
counts_modified = result.get_counts()
print(counts_modified)
plot_histogram(counts_modified); plt.show()
```

3. Expected comparison

Circuit	Expected outcomes	Approx. counts (100 shots)	Correlation
Original Bell	00 and 11	About 50 each	Same outcomes
Modified with X	01 and 10	About 50 each	Opposite outcomes

4. Explanation and conclusion

The original circuit prepares the Bell state (|00■ + |11■)/√2, so the two qubits produce the same measurement result. Applying one X gate to qubit 0 flips that qubit and transforms the state into (|10■ + |01■)/√2. The measurement outcomes become opposite. Therefore, changing one gate changes the relationship from correlated to anti-correlated. In an ideal simulator, each allowed outcome has probability 50%; actual counts can vary slightly with 100 shots.

Note: Qiskit displays classical bit 0 on the right of count strings. The exact printed labels depend on the measurement mapping. Qiskit was not installed in the preparation environment, so the supplied code generates the actual counts and histograms when run.